
UK Space-Based Solar Power Cost-Condition Map

Report subtitle: A reproducible threshold assessment for UK decision screening — not a deployment forecast

Evidence status: source registry reviewed to 2026-07-19; selected official DESNZ benchmark cells are machine-reconciled.

Version: v1.2

Prepared as: a bilingual, executable techno-economic research artifact.

Disclaimer: this report identifies model-conditional cost requirements. It does not demonstrate engineering feasibility, predict a commissioning year, estimate an investment return or recommend an investment.

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1. Executive answer

The short answer is: **the reference configuration is expensive, but the project is useful because it reveals the conditions behind that result.** At the study-authored reference point, grid-connection-point LCOE is **£429/MWh**. This is a normalization anchor, not a forecast or preferred design.

Table 1. Study-defined cost lines and their meaning

Cost line	Use in this project	Decision meaning and limitation
£150/MWh	Broad study-defined screening line	A model diagnostic, not an official UK threshold
£100–120/MWh	Study-defined screen for a potential firm or near-firm role	Triggers deeper engineering, finance and system-value work
£80/MWh	Overlap with the selected system-adjusted comparator	Still not a market-competitiveness conclusion
£60/MWh	Stringent stress test	No tested one-way input reaches it; coupled cases remain exploratory

The strongest one-way result is specific mass, **within this model boundary and the selected ranges, with every other input held at its reference value.** The limiting values are 1.28, 0.88 and 0.62 kg/kW-space for £150, £120 and £100/MWh respectively. No one-way specific-mass value in the explored range reaches £80 or £60/MWh.

Launch cost alone is not sufficient: even at the favourable explored bound of £20/kg, LCOE remains about £204/MWh. Raising end-to-end efficiency alone to 35% leaves about £188/MWh. The model therefore points to testing multiple coupled combinations across mass, launch, space hardware, assembly, finance, efficiency and delivered energy; it does not prove that every variable must move.

The contribution is **not greater forecasting accuracy.** It is a transparent and reproducible answer to a narrower question: what cost and performance conditions would have to hold, inside a stated model, before SBSP enters selected UK decision regions?

2. Research question and distinctive contribution

The representative SBSP studies reviewed here evaluate named architectures, future deployment cases, lifecycle impacts or system pathways. This project complements them with continuous one-way and coupled parameter frontiers. It therefore makes assumptions, conditional thresholds and failure-to-reach results directly auditable.

Table 2. Relationship to selected existing SBSP studies

Study	Primary analysis	Reported context	Relationship to this project
UK SBSP Phase 2 economic study (2021)	architecture and future scenario study	£35-79/MWh (p10-p90) with p50 about £50/MWh in 2018 prices; 2 GW CASSIOPeiA; 30-year life; 20% hurdle; 2040 commissioning case	Provides architecture and scenario context; this project instead exposes continuous model-conditional cost thresholds.
UK small-scale SBSP study (completed 2025; published 2026)	architecture, pathways and system-value study	2024 GBP: £335-595/MWh (2030), £154-249/MWh (2035), £87-129/MWh (2040)	Provides a current external consistency check and architecture-specific drivers; this project adds an auditable parameter-space map.
NASA OTPS assessment (2024)	lifecycle cost and emissions assessment	FY2022 USD: \$610/\$1,590 per MWh baseline for RD1/RD2; \$30/\$80 per MWh under combined favourable assumptions	Shows a roughly 20-fold lifecycle-cost difference between NASA's selected baseline and favourable-combination cases; no NASA value is imported into this model.
This project (v1.2)	reproducible continuous cost-condition assessment	£429/MWh study-authored reference anchor; root-solved conditional lines at £150/120/100/80/60 per MWh	Distinctive contribution: traceable inputs, continuous one-way and coupled frontiers, bilingual outputs and executable tests.

The project's publishable advantage is specific:

- Every exact model input is classified as a **study-authored exploratory assumption**; literature values are kept in a separate evidence map.
- Thresholds are solved by high-precision monotonic bisection rather than read from a coarse plotting grid.
- The analysis reports both attainable and unattainable one-way targets across 16 variables and 5 study-defined cost lines.
- Coupled frontiers and 3 alternative parameter slices show that the answer is not a single-variable launch-cost story.

- Source files, processed outputs, figures, English and Chinese reports and automated checks are regenerated by one command.

These features make the analysis **more transparent for threshold questions**, not universally more accurate than architecture-specific studies.

3. Scope, accounting boundary and method

The model ends at the **grid connection point**. It includes space hardware, wireless-power hardware, launch, orbit transfer, in-orbit assembly, rectenna, local grid connection, programme margin, capital recovery, fixed OPEX, refurbishment and variable OPEX. It excludes downstream transmission reinforcement, balancing, storage, curtailment, reliability services, taxes, decommissioning cost, revenue design and monetised system value.

All model financial inputs are interpreted in 2024 real GBP. The wacc variable is a **real project discount-rate proxy** used in the capital-recovery factor; tax, inflation and financing tranches are outside the model. The delivered capacity factor is also a proxy combining availability, outages and operational constraints; it is not an observed SBSP availability statistic.

The calculation is:

- annual delivered energy = delivered capacity × 8,760 hours × delivered capacity factor;
- required space power = delivered grid capacity ÷ end-to-end efficiency;
- orbital mass = required space power × whole-architecture specific mass;
- initial CAPEX = summed component CAPEX × (1 + programme margin);
- LCOE = (annualised CAPEX + fixed OPEX + refurbishment + variable OPEX) ÷ annual delivered MWh.

A one-way threshold changes one input and fixes all others. A coupled equal-fraction frontier moves selected inputs from the reference point toward their favourable recorded bounds by the same normalized fraction. That fraction is a mathematical index, **not technology readiness, probability, calendar progress or a unique roadmap**.

4. Evidence quality and source discipline

The evidence system separates three concepts:

- **Direct input:** an exact value used in the model. All SBSP reference values and ranges are study-authored assumptions.
- **Contextual evidence:** published metrics showing plausible scales or important drivers without being treated as equivalent definitions.
- **External comparison:** results from a different architecture or boundary, used to test narrative consistency rather than validate this model.

The withdrawn 2020 UK Space Agency press release is retained only as programme history. It supplies no numerical input. Caltech's 160 g/m² areal density is not converted into whole-system kg/kW-space. Likewise, the official small-scale study completed in 2025 and published in 2026 supplies only external context: its specific power, hurdle rates and high-Earth-orbit launch estimates use different definitions.

The £429/MWh reference anchor lies inside the 2026-published official small-scale study's reported £335–595/MWh 2030 band. This is a useful consistency observation, **not validation**, because scale, architecture, orbit, scenario year, financing and cost boundaries differ.

5. Reference anchor and cost structure

Table 3. Core reference inputs

Parameter	Reference value	Role in the model
Delivered grid capacity	2,000 MW	Scale anchor; most unit costs scale with capacity
End-to-end efficiency	15.0%	Sets required space-side power and therefore mass
Delivered capacity factor (model proxy)	90.0%	Delivered-energy proxy combining availability and constraints
Specific mass	5.00 kg/kW-space	Scales launch, transfer and assembly together
Launch cost	£500/kg to staging orbit	Transport-cost lever to the model-defined staging orbit
Space hardware cost	£0.40/W-space	Orbital manufacturing-cost lever
In-orbit assembly and deployment cost	£200/kg	Mass-scaled deployment allowance
Real project discount-rate proxy	6.5%	Real project discount-rate proxy used in capital recovery

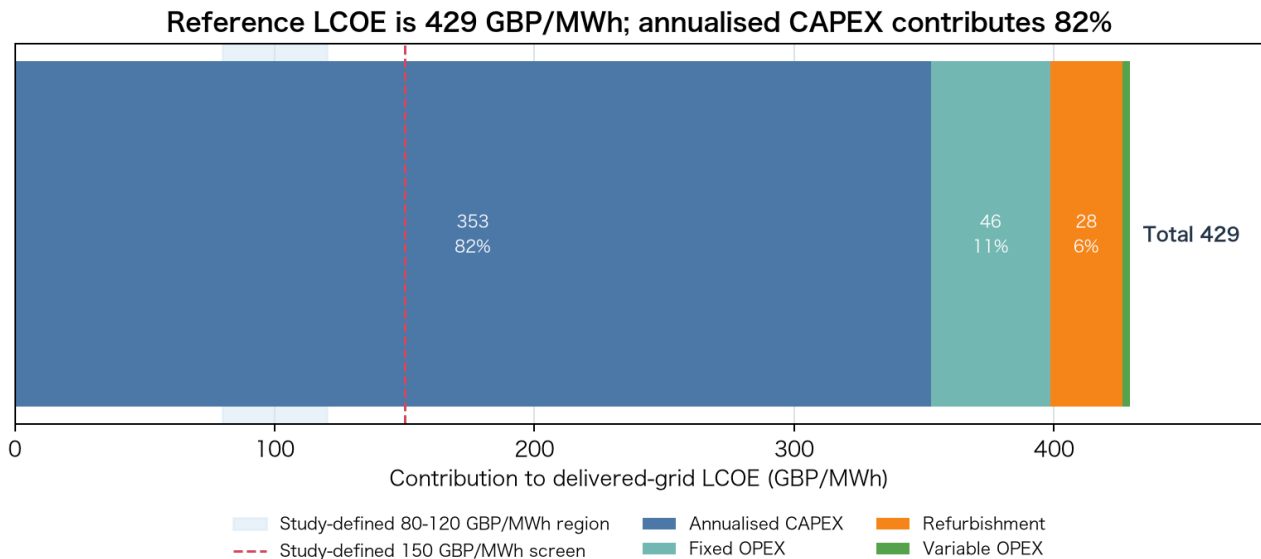
Parameter	Reference value	Role in the model
System lifetime	30 years	Period over which capital is recovered

The 2.0 GW delivered-capacity anchor requires 13.33 GW of space-side power and **66.67 million kg** of orbital hardware, or about 66,667 tonnes. Both units are shown to make the mass scale unambiguous.

Table 4. Reference initial CAPEX reconciliation

Component	Cost
Launch	£33.33bn
In-orbit assembly and deployment	£13.33bn
Orbit transfer	£6.67bn
Space-segment hardware	£5.33bn
Wireless-power hardware	£1.33bn
Rectenna	£0.30bn
Grid connection	£0.20bn
Pre-margin subtotal	£60.50bn
Programme margin	£12.10bn
Initial CAPEX	£72.60bn

Figure 1. Reference LCOE components

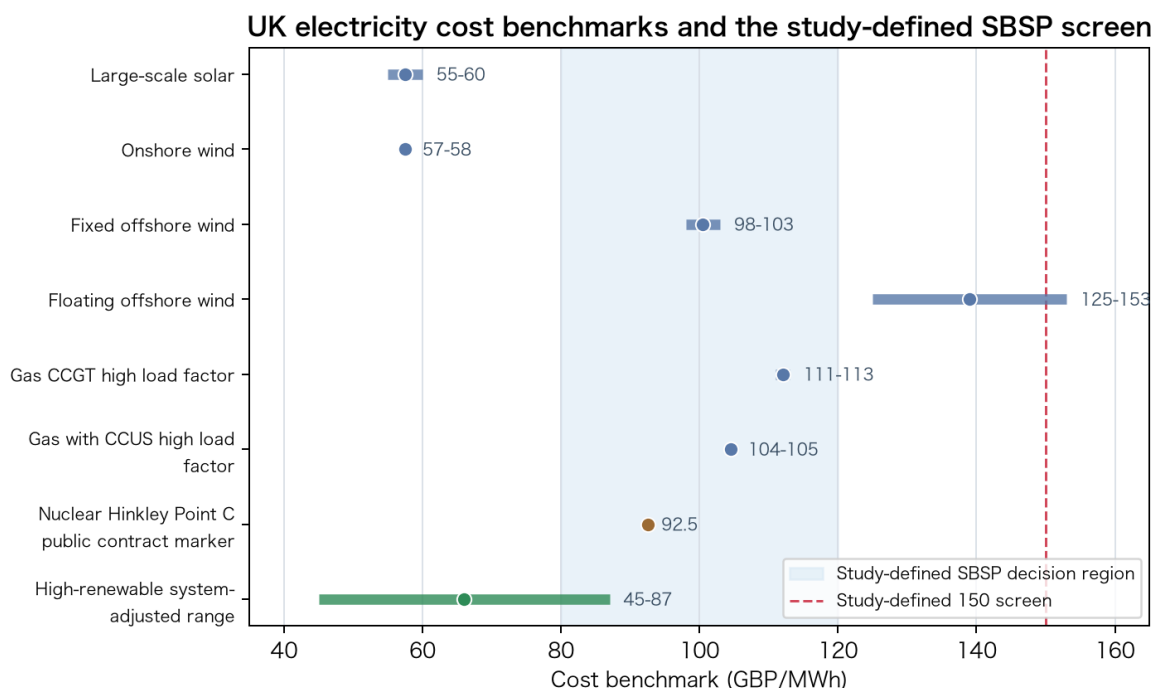


Interpretation. Annualised CAPEX contributes about 82% of annual reference cost. The result is therefore especially sensitive to physical scale and capital recovery.

The reference point produces 15.768 million MWh/year, £72.6bn initial CAPEX and £6.77bn annualised total cost. These figures reconcile exactly in the generated model outputs.

6. UK comparator landscape

Figure 2. UK electricity cost comparators



DESNZ values are 2024 real GBP; BEIS system-adjusted values are 2018 real GBP; the Hinkley CfD marker is in 2012 prices and CPI-indexed.

Interpretation. The figure shows the generation-only rows flagged for the headline band, plus the Hinkley contract marker and the historical system-adjusted band. The plotted generation rows span about £55–£153/MWh; excluding the floating-offshore FOAK row gives £55–£113/MWh.

Table 5. Comparator definitions from the structured data

Comparator	Reported value	Price basis	Metric type
Large-scale solar	£55–60/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Onshore wind	£57–58/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Fixed offshore wind	£98–103/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Floating offshore wind	£125–153/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Gas CCGT high load factor	£111–113/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Gas with CCUS high load factor	£104–105/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Gas with CCUS mid load factor	£181/MWh	2024 real GBP	DESNZ 2025 Annex A 2030-2035 range
Nuclear Hinkley Point C public contract marker	£92.5/MWh	2012 real-price terms (CPI-indexed)	35-year Contract for Difference strike-price marker
High-renewable system-adjusted band	£45–87/MWh	2018 real GBP	Indicative enhanced-LCOE envelope

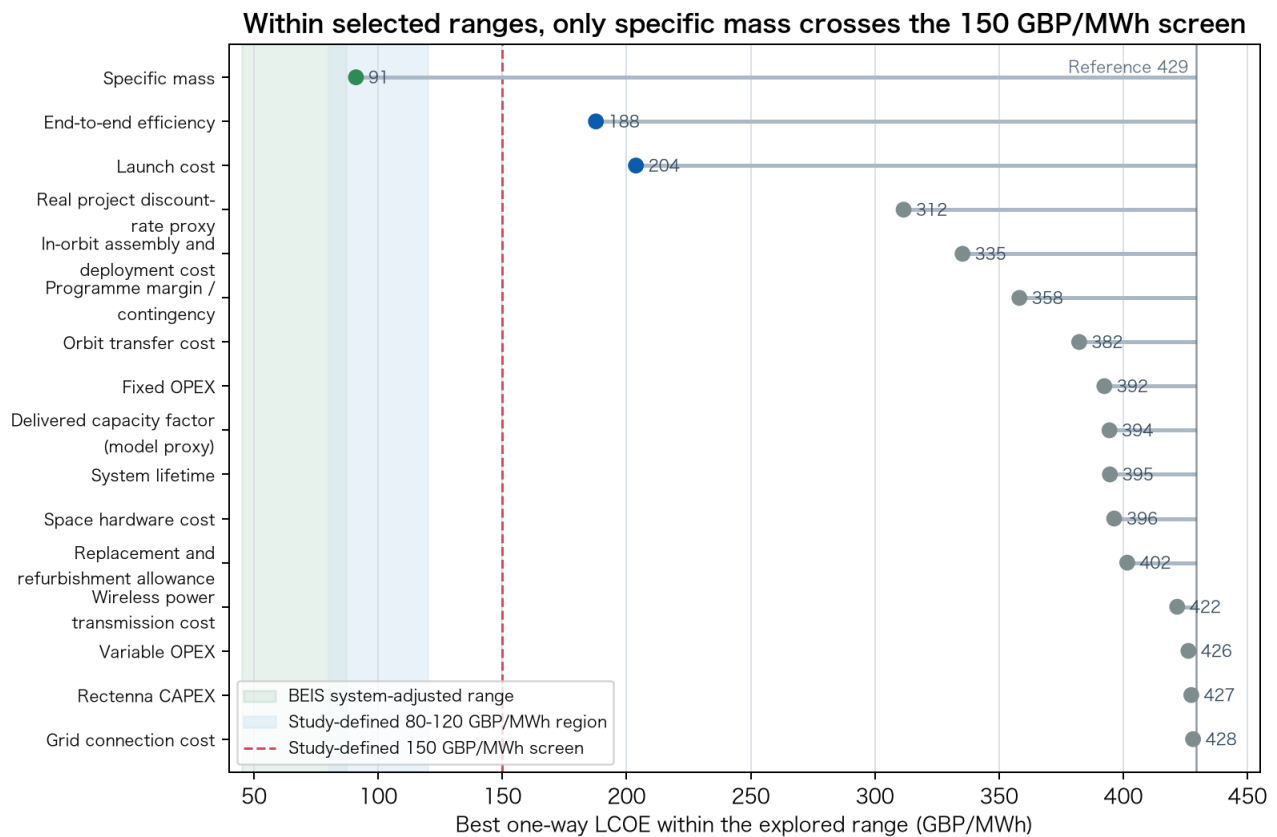
The Hinkley Point C £92.50/MWh marker is a 35-year CfD strike price stated in 2012 prices and CPI-indexed; it is not a generic nuclear LCOE. The BEIS enhanced-LCOE range of £45–£87/MWh is in 2018 real GBP, whereas the DESNZ generation rows are in 2024 real GBP. No false-precision price conversion is applied.

Figure 2 uses the six generation-only rows marked for the headline band. Table 5 additionally retains the £181/MWh mid-load-factor gas-CCUS sensitivity, but it is excluded from the figure and headline band to avoid counting two utilisation cases for the same technology as separate headline comparators.

Consequently, overlap with a chart band is a screening result only. Wholesale prices, contract prices, generator LCOE and system-adjusted costs are different metrics.

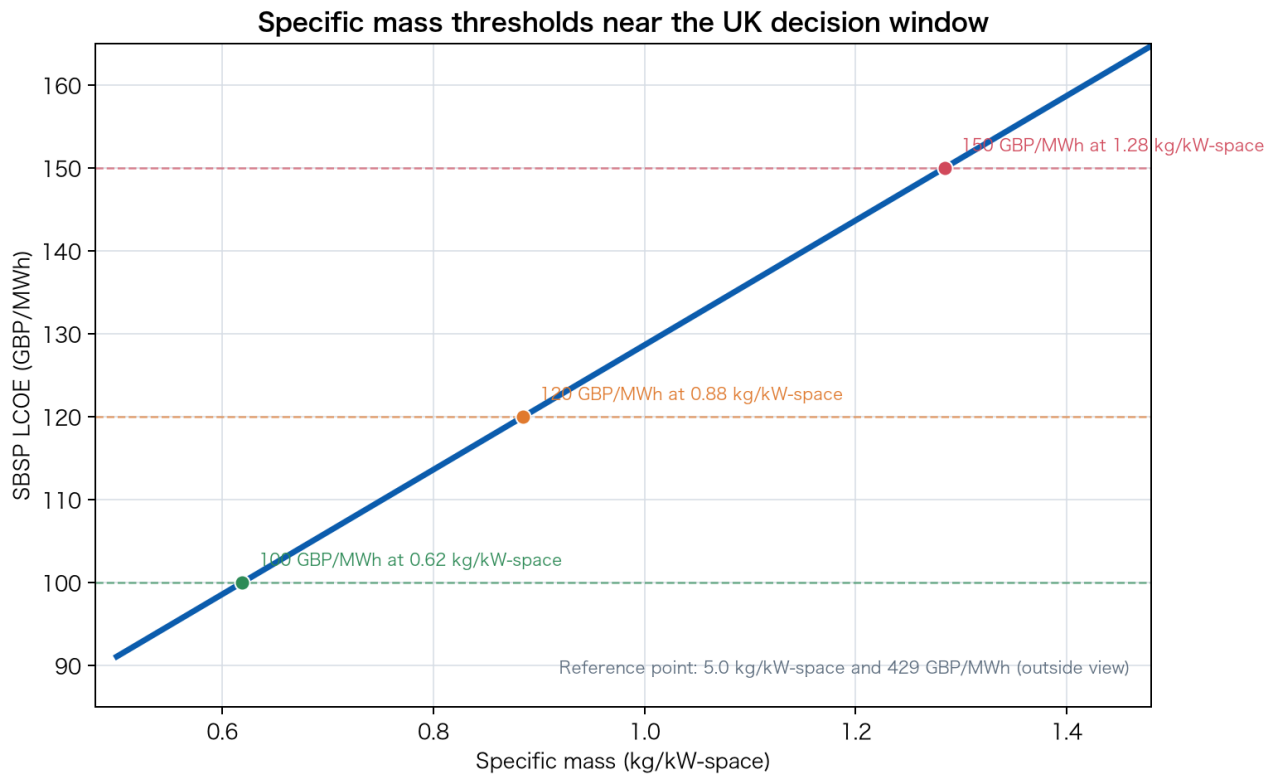
7. One-way conditional thresholds

Figure 3. Best LCOE attainable one variable at a time



Interpretation. Within the selected one-way ranges, only whole-architecture specific mass crosses the £150/MWh study line. This ranking is range-dependent and is not universal across architectures.

Figure 4. Specific-mass thresholds in the decision window



Interpretation. With all other reference inputs fixed, the model requires at most 1.28, 0.88 and 0.62 kg/kW-space for £150, £120 and £100/MWh. These are conditional mathematical requirements, not proven engineering targets.

Table 6. Complete one-way threshold audit

Input	Unit	£150	£120	£100	£80	£60
Launch cost	GBP/kg to staging orbit	—	—	—	—	—
Specific mass	kg/kW-space	1.28	0.88	0.62	—	—
Space hardware cost	GBP/W-space	—	—	—	—	—
Wireless power transmission cost	GBP/W-space	—	—	—	—	—
End-to-end efficiency	fraction	—	—	—	—	—
Real project discount-rate proxy	fraction	—	—	—	—	—
Rectenna CAPEX	GBP/W-delivered	—	—	—	—	—
Grid connection cost	GBP/kW-delivered	—	—	—	—	—
System lifetime	years	—	—	—	—	—
Delivered capacity factor (model proxy)	fraction	—	—	—	—	—
In-orbit assembly and deployment cost	GBP/kg	—	—	—	—	—
Orbit transfer cost	GBP/kg	—	—	—	—	—
Programme margin / contingency	fraction	—	—	—	—	—

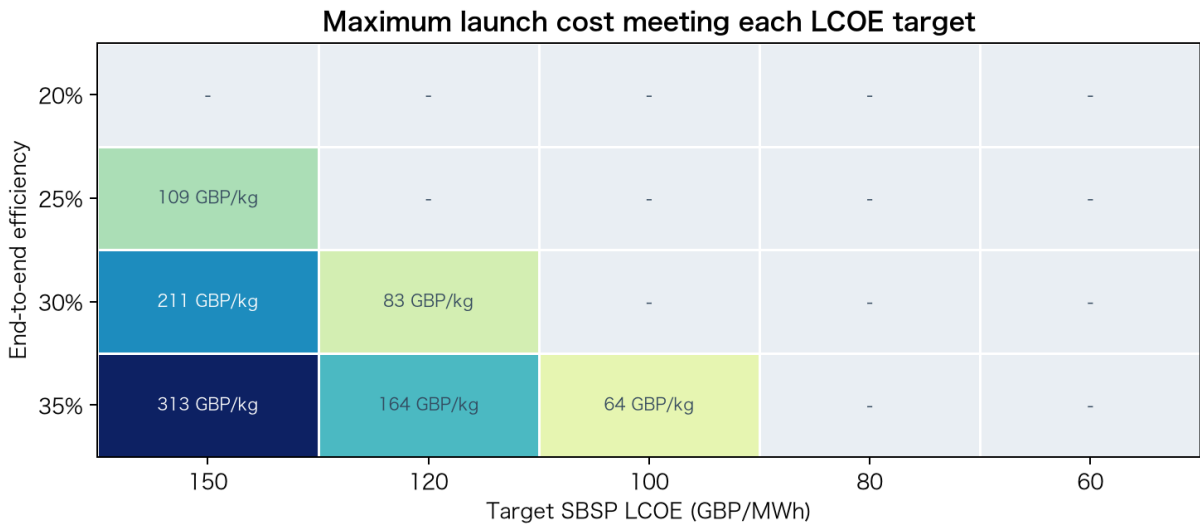
Input	Unit	£150	£120	£100	£80	£60
Replacement and refurbishment allowance	fraction/year	—	—	—	—	—
Fixed OPEX	fraction/year	—	—	—	—	—
Variable OPEX	GBP/MWh	—	—	—	—	—

An em dash means that no value reaches the target inside the tested one-way bound; it does not mean physical impossibility outside that bound.

The result does not mean that specific mass is the only necessary condition. It means only that, under the reference assumptions and selected one-way bounds, it is the sole individual variable able to cross £150/MWh. In a coupled design, low mass is not sufficient for every target, and other architectures may rank drivers differently. The official small-scale study published in 2026, for example, attributes 55.5–64.0% of its own LCOE variance to launch assumptions.

8. Coupled cost-condition frontiers

Figure 5. Launch-cost and efficiency frontier



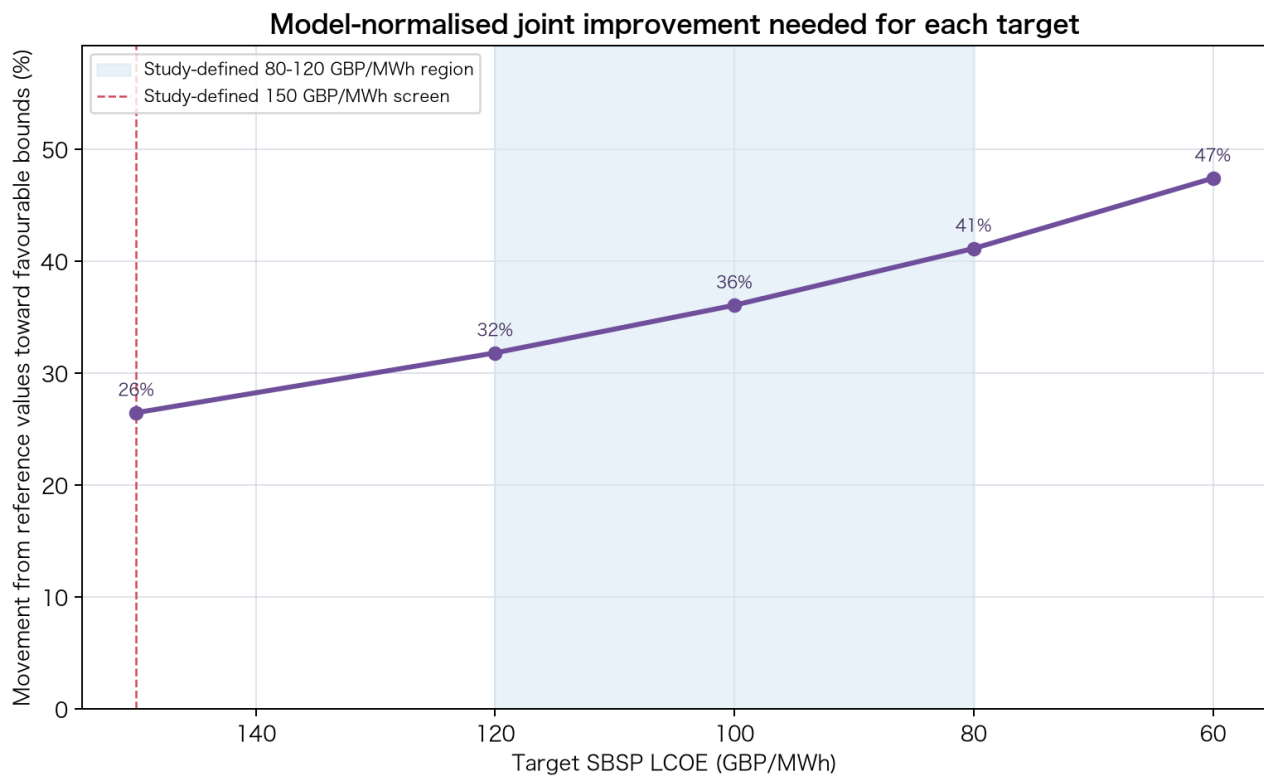
A dash means no solution within the explored 20-5,000 GBP/kg launch-cost range.

Interpretation. Higher efficiency relaxes the launch-cost condition, but many cells remain unattainable when the remaining reference inputs are fixed. Values are root-solved inside the recorded launch-cost bounds.

Table 7. Selected launch-cost limits with all other inputs fixed

End-to-end efficiency	Target	Maximum launch cost
25%	£150/MWh	£109.1/kg or lower
25%	£120/MWh	not reached
25%	£100/MWh	not reached
30%	£150/MWh	£210.9/kg or lower
30%	£120/MWh	£83.2/kg or lower
30%	£100/MWh	not reached
35%	£150/MWh	£312.8/kg or lower
35%	£120/MWh	£163.7/kg or lower
35%	£100/MWh	£64.4/kg or lower

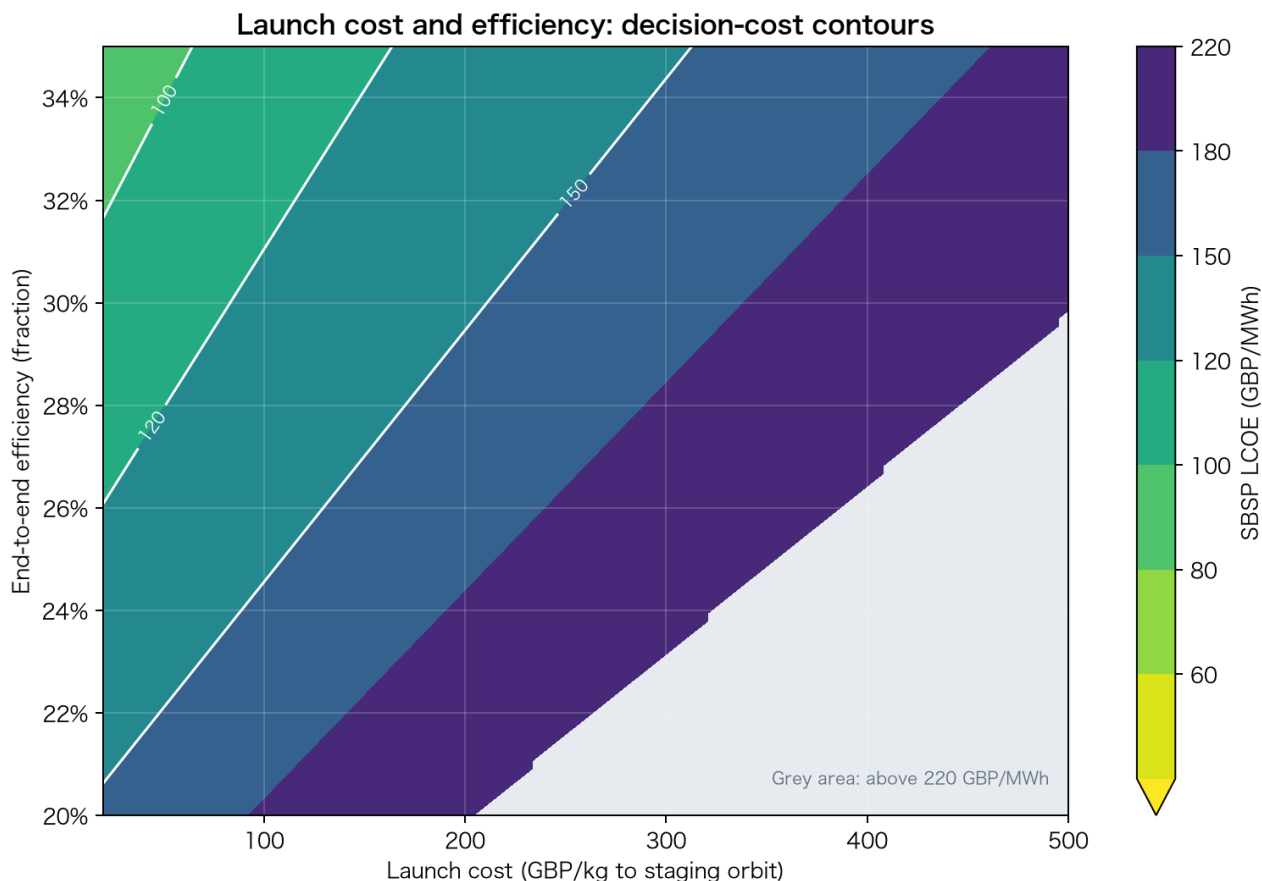
Figure 6. Equal-fraction coupled frontier



Interpretation. Lower target LCOEs on the x-axis correspond to larger normalized movements on the y-axis. The y-axis is a mathematical interpolation index, not an implementation timeline.

Table 8. Coupled equal-fraction index and selected parameter values

Target	Normalized movement	Launch	Specific mass	Efficiency	Real discount proxy
£150/MWh	26.5%	£373/kg	3.81 kg/kW-space	20.3%	5.57%
£120/MWh	31.8%	£347/kg	3.57 kg/kW-space	21.4%	5.39%
£100/MWh	36.1%	£327/kg	3.38 kg/kW-space	22.2%	5.24%
£80/MWh	41.1%	£303/kg	3.15 kg/kW-space	23.2%	5.06%
£60/MWh	47.4%	£272/kg	2.87 kg/kW-space	24.5%	4.84%

Figure 7. Launch cost $\times$ efficiency decision contour

Interpretation. The two-variable contour makes interaction visible, but it still fixes every unplotted input at the reference point. It must not be read as a complete design feasibility map.

The equal-fraction results are one transparent index among many possible paths. The generated alternative slices show that high efficiency, low mass or infrastructure-like finance can each reshape the frontier. None is presented as the unique or necessary roadmap.

9. Interpretation for UK decisions

The practical reading is sequential:

- Above £150/MWh, the model remains outside its broad study-defined screening line; research should focus on whether integrated architecture evidence can change several major drivers together.
- At £100–120/MWh, a credible integrated design would justify detailed engineering, financing and GB dispatch/network analysis.
- At £80/MWh or below, cost overlap with the selected historical system-adjusted band becomes more relevant, but differing price bases and omitted system effects still prevent a market conclusion.
- Any claim of firm low-carbon value must be tested in a full GB power-system model rather than subtracted informally from plant LCOE.

The official small-scale study published in 2026 reports a £21/MWh system-benefit adjustment in each of its cases. This report does not subtract that figure because the architecture and system-model boundary differ.

10. Limitations, robustness and non-claims

The strongest robustness feature is traceability: official 2025 generation-cost cells are checked directly against the stored workbook; source IDs are foreign-key validated; input CSVs are structurally validated; model accounting reconciles; and threshold roots are tested against their target LCOEs.

The principal limitations are:

- Architecture feasibility is not assessed: beam safety, spectrum, thermal control, degradation, debris risk, deployment and maintainability require engineering models and demonstrations.
- Parameter ranges are deliberately broad and partly judgmental; driver rankings depend on those ranges and the reference point.
- Correlations, construction schedules, learning curves, probabilistic uncertainty and tax/financing structure are not modelled.
- The grid-connection-point boundary omits downstream network, balancing, storage, curtailment, reliability and market effects.
- Comparator price bases are disclosed but not harmonised.
- External SBSP studies are not like-for-like validation datasets.

This project does **not** claim to be the first threshold analysis, to discover universal necessary conditions, or to produce a more accurate deployment forecast. Its defensible special feature is an openly executable, bilingual and source-disciplined cost-condition map.

11. Conclusion

At the study-authored reference point, SBSP LCOE is approximately £429/MWh at the grid connection point. Lower launch cost by itself does not close the gap. Within the selected one-way ranges, a very low whole-architecture specific mass is the only individual lever that reaches £150/MWh, and it still cannot reach £80/MWh alone.

The central conclusion is therefore narrower and conditional: **lower launch cost or higher efficiency alone is insufficient at the tested bounds; no single tested one-way change reaches £80 or £60/MWh, so some coupled improvement is required for those lines.** The analysis does not prove that every listed variable must improve, or that its equal-fraction path is unique. It also does not prove that any engineering or commercial combination can be achieved.

The project's value is the clarity of that conditional statement and the audit trail behind it.

Appendix A. Evidence-to-claim map

Table A1. Contextual evidence and its limitations

Parameter or claim	Evidence role / source	Locator	Published context	Applicability limit
delivered_capacity_mw	external precedent / UK_SBSP_2021_P HASE2	PDF pp.18-19	2 GW CASSIOPeiA case	The same number is used here as a study scale anchor, not as evidence that this architecture is preferred.
system_lifetime_years	external precedent / UK_SBSP_2021_P HASE2	PDF pp.18-19	30 years; the later small-scale study uses 15 years	Lifetime remains architecture-dependent and the model does not demonstrate survivability or refurbishment feasibility.
end_to_end_efficiency	technical context / C ALTECH_SBSP_2022	abstract	7-14% end-to-end efficiency	Concept and subsystem boundaries differ; this does not directly set the model's 15% reference value.
end_to_end_efficiency	source-inspired assumption / UK_SBSP_2021_ANNEX_B	PDF p.4	approximately 14.85-24.39% from independent factor products	The 15% reference is still a project assumption; multiplying independent extrema is not a probability distribution or validated integrated design.
end_to_end_efficiency	external consistency check / DESNZ_SBSP_2025	Table 1, PDF pp.11-12; reference-design selection p.13	11.7-19.9%; selected Space Solar small-scale reference design 14.9%	The report states that major values are manufacturer-claimed, not independently verified; missing values are inferred and definitions may differ.
capacity_factor	scope qualification / DESNZ_SBSP_2025	Table 3, PDF p.16	HEO: UK rectenna 95.7%, satellite average 53.9%; circular LEO: satellite average 20.1%, UK rectenna 21.5%	Rectenna utilization is not equivalent to this model's delivered capacity factor; the 90% value remains a study assumption.
specific_mass_kg_per_kw_space_power	technical context / C ALTECH_SBSP_2022	abstract	160 g/m2 areal density	Areal density cannot be converted to whole-system kg/kW-space without power density and complete subsystem boundaries.
specific_mass_kg_per_kw_space_power	external consistency check / DESNZ_SBSP_2025	Table 1, PDF pp.11-12; reference-design selection p.13	0.15-0.67 kW/kg; selected Space Solar small-scale reference design 0.58 kW/kg	Values are architecture-specific and partly manufacturer-claimed; inverse conversion is not imported as the model's whole-system specific mass.
launch_cost_gbp_per_kg	external range / UK_SBSP_2021_ANNEX_B	PDF p.5 and p.11	£358-2,410/kg	Orbit, procurement, price year and service boundary differ from the model-defined staging-orbit variable.
launch_cost_gbp_per_kg	external comparison / DESNZ_SBSP_2025	Table 12 p.37; Tables 27-28 pp.71-73	central HEO £1,647/1,400/1,153 per kg; optimistic £1,188/1,008/827 for 2030/35/40	Destination orbit, scenario year, vehicle, procurement and service boundary differ; although both are stated in 2024 GBP, the model's £500/kg reference is not taken from this table.

Parameter or claim	Evidence role / source	Locator	Published context	Applicability limit
launch_cost_gbp_per_kg	external precedent / NASA_OTPS_SBSP_2024	PDF p.10	nominal \$500/kg; \$425/kg after a 15% quantity discount	FY2022 USD, US procurement and architecture boundaries differ; no currency conversion or direct input transfer is made.
wacc	financing context / DESNZ_SBSP_2025	Table 12 p.37; Table 26 pp.68-69	20%/13.2%/9.06% scenario hurdles; 14.9%, 9.8%, 6.75%, 4.83% investor-class cases	Hurdle rate is not identical to WACC; this model's 6.5% real discount proxy remains study-authored.
wacc	source-inspired assumption / DESNZ_EGC_2025_ANNEX_A	Technical and Cost Assumptions, row 38	6.5% for large solar and onshore wind	This is not an observed SBSP WACC and does not specify this model's tax or financing structure.
grid_connection_cost_gbp_per_kw_delivered	source-inspired assumption / DESNZ_SBSP_2025	PDF p.70	£0.15m/MW = £150/kW	The model uses £100/kW as its own simplified allowance; scope and site conditions may differ.
fixed_opex_pct_capex_per_year	source-inspired assumption / UK_SBSP_2021_ANNEX_B	Table B1, PDF p.4	triangular lower/mode/upper values of 0.8%/1.9%/4.7%	Definitions and cost bases vary; the model's 1% of CAPEX per year remains a study assumption.
reference_case_lcoe	external consistency check / DESNZ_SBSP_2025	executive summary, PDF pp.3-4	2024 GBP: £335-595/MWh (2030), £154-249 (2035), £87-129 (2040)	This is not validation: scale, architecture, orbit, scenario year, financing and accounting boundaries differ.
competitive_cost_claim	external comparison / UK_SBSP_2021_P_HASE2	Executive summary p.3; assumptions pp.18-19; Figure 5 p.22 and Table 4 p.23	£35-79/MWh (p10-p90), p50 about £50/MWh, 2018 prices; 2 GW; 30 years; 20% hurdle	Architecture, learning, commissioning year, financing and price basis differ; values are not ranked as like-for-like forecasts.
architecture_dependence	external comparison / NASA_OTPS_SBSP_2024	executive summary pp.3-4; pp.7-10	FY2022 USD: \$610/\$1,590 per MWh baseline; \$30/\$80 per MWh favourable combination	US architectures, service profiles, currency and lifecycle boundary differ; no NASA value is imported into this model.
launch_cost_importance	scope qualification / DESNZ_SBSP_2025	Table 14, PDF p.38	55.5%/62.9%/64.0% for 2030/35/40	This project may rank specific mass first within its own one-way ranges; neither ranking is universal.
system_value	system context / DESNZ_SBSP_2025	executive summary, PDF pp.3-4	adjusting for system benefits reduces LCOE by £21/MWh in all cases	The present model does not monetize whole-system value; £21/MWh is not subtracted from project LCOE.
system_adjusted_comparison	benchmark definition / BEIS_EGC_2020	Tables 7.1-7.3	transcribed ranges stored in uk_system_justified_costs.csv	2018 real GBP and scenario-specific; indicative only and not directly harmonised with the model's 2024 real GBP basis.

Appendix B. Source registry

Table B1. Sources used by the project

Source ID	Reference	Organisation	Role
DESNZ_EGC_2025	Electricity generation costs 2025	Department for Energy Security and Net Zero	UK generation benchmark source
DESNZ_EGC_2025_ANNEX_A	Annex A: Additional estimates and key assumptions 2025	Department for Energy Security and Net Zero	UK generation benchmark input workbook
DESNZ_EGC_2023	Electricity generation costs 2023	Department for Energy Security and Net Zero	Supplementary UK benchmark source
BEIS_EGC_2020	BEIS Electricity Generation Costs (2020)	Department for Business, Energy and Industrial Strategy	System-adjusted benchmark source
OFGEM_WHOLESALE	Wholesale market indicators	Ofgem	Market context
NESO_BALANCING_2025	2025 Annual Balancing Costs Report	National Energy System Operator	System-cost evidence
NESO_NETWORK_UPDATE_2026	Beyond 2030 - Electricity Transmission Update	National Energy System Operator	System-cost evidence
NESO_CP2030	Advice on achieving clean power by 2030	National Energy System Operator	System-cost evidence
NESO_OPERABILITY_2026	Operability Strategy Report and Electricity Markets Roadmap	National Energy System Operator	System-cost evidence
UKSA_SBSP_2020	UK government commissions space solar power stations research	UK Space Agency and BEIS	Historical SBSP context
UK_SBSP_2021	Space based solar power: de-risking the pathway to net zero	Department for Business, Energy and Industrial Strategy	External SBSP study landing page
UK_SBSP_2021_PHASE2	Space Based Solar Power as a Contributor to Net Zero — Phase 2: Economic Feasibility	Frazer-Nash Consultancy for BEIS	External SBSP economic study
UK_SBSP_2021_ANNEX_B	Space Based Solar Power as a Contributor to Net Zero — Phase 2: Economic Feasibility – Annex B: Input Data Sources	Frazer-Nash Consultancy for BEIS	External SBSP input evidence
DESNZ_SBSP_2025	Feasibility of Small-Scale Space Based Solar Power (SBSP) Systems for Early Market Adoption	Frazer-Nash Consultancy; Space Solar Engineering Ltd; Imperial College London for DESNZ	External SBSP study
NASA_OTPS_SBSP_2024	Space Based Solar Power	NASA Office of Technology, Policy, and Strategy	External SBSP study
CALTECH_SBSP_2022	A Lightweight Space-based Solar Power Generation and Transmission Satellite	Caltech Space Solar Power Project	SBSP technical context
CALTECH_WPT_2024	Wireless Power Transfer in Space using Flexible, Lightweight, Coherent Arrays	Caltech Space Solar Power Project	SBSP technical context
NAO_HPC_2017	Hinkley Point C	National Audit Office	Nuclear benchmark context
ASSUMPTION_THIS_STUDY	Exploratory SBSP cost-condition model assumptions	This project	Model assumption source

Appendix C. Assumptions and boundary register

Table C1. Explicit analytical assumptions

ID	Area	Assumption	Limitation
A1	Analytical framing	This is a cost-threshold assessment. No SBSP result is expressed as a deployment-year forecast.	Official UK benchmarks may contain source publication years or commissioning references, but they are not used as SBSP scenarios.
A2	Unit and price convention	All model financial inputs and outputs are interpreted in 2024 real GBP and LCOE is reported in GBP/MWh at the grid connection point.	Some external benchmarks use different real-price bases; these are labelled rather than silently harmonised.
A3	Efficiency boundary	End-to-end efficiency is applied by increasing required space-side power capacity for a fixed delivered-grid capacity.	This simplifies subsystem losses into one continuous variable.
A4	Mass boundary	Specific mass is applied per kW of required space-side power capacity.	Architecture-specific mass allocation between arrays, structures and transmitters is not modelled.
A5	Contingency	Programme margin is applied to all pre-margin CAPEX categories.	The model does not separately estimate owner costs, insurance during construction or first-of-a-kind programme development.
A6	Refurbishment	Replacement and refurbishment are modelled as an annual allowance based on initial CAPEX.	Periodic major replacement timing is not explicitly optimised.
A7	System costs	Renewable system-adjusted benchmarks use BEIS enhanced LCOE ranges from 2020 because the latest DESNZ 2025 LCOE report says wider system costs require full system modelling.	Values are in 2018 real GBP and should be treated as indicative ranges.
A8	Nuclear	Nuclear is represented only by a public contract-price marker because recent DESNZ reports do not publish an updated generic nuclear LCOE.	Not directly comparable with 2024 real generation LCOE.
A9	Conditional thresholds	A threshold is necessary only within this model, its accounting boundary and the stated parameter ranges; it is not a universal engineering requirement.	Alternative architectures, coupled parameter changes or omitted system value can move the threshold.
A10	External comparisons	Published SBSP cost estimates are used as context and consistency checks, not as validation or direct substitutes for this model's inputs.	Architectures, orbit destinations, price years, financing and cost boundaries differ materially across studies.
A11	Discount rate	The parameter named wacc is implemented as a real project discount-rate proxy in the capital-recovery factor; taxes, inflation and financing tranches are not modelled.	It should not be read as a fully specified observed pre-tax or post-tax corporate WACC.
A12	Delivered-energy boundary	The model ends at the grid connection point and uses a delivered capacity-factor proxy that combines availability, outages and operating constraints.	Downstream transmission reinforcement, balancing, storage, reliability services, curtailment and monetised system value are excluded.

Machine-readable inputs and all complete sensitivity curves are retained under data/; the full English figure set is under figures/; exact generated thresholds are under data/processed/.